LangChain — Overview & Architecture

Section 00: Why LangChain Exists and How It Fits Together

LangChain v1.2.x Study Guide

March 2026

What You Will Learn

Section Objectives

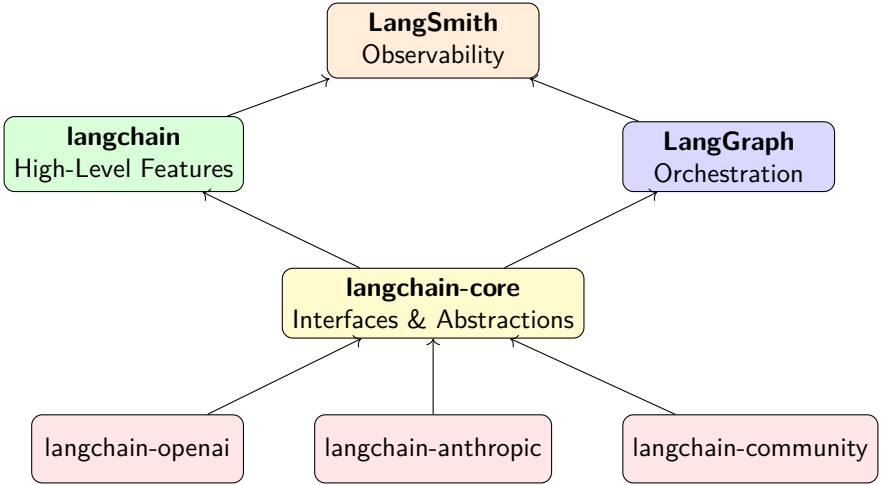
By the end of this section, you will understand:

- What problem LangChain solves and why it exists
- The LangChain ecosystem: LangChain, LangGraph, LangSmith
- The package architecture and how modules relate
- How to install and configure LangChain
- Where LangChain fits relative to alternatives

Prerequisites: Basic Python knowledge. No prior LangChain experience needed.

[Docs: langchain.com/docs]

Mental Model: The LangChain Universe



[Docs: docs.langchain.com/oss/python/concepts]

The Problem: Why LangChain?

Building LLM applications without a framework means:

- Writing custom glue code for every LLM provider (OpenAI, Anthropic, Google, ...)
- Manually managing prompt templates, output parsing, error handling
- Reinventing conversation memory, tool calling, retrieval patterns
- No standardized way to compose, test, or observe your pipelines

LangChain's Value Proposition

LangChain provides **standardized interfaces** for LLM components and **composable primitives** for chaining them together — so you can swap providers, add tools, and build agents without rewriting your application.

[Docs: langchain.com]

What LangChain Is (and Isn't)

LangChain IS

- A framework for composing LLM calls
- An abstraction layer over providers
- A toolkit for agents, RAG, tools
- An orchestration platform (via LangGraph)
- Provider-agnostic by design

LangChain is NOT

- An LLM itself
- A hosted AI service
- A UI framework
- Required for all LLM apps
- A replacement for understanding LLMs

When to Use LangChain

Use LangChain when you need **multi-step LLM workflows**, **tool calling**, **re-**

The Ecosystem at a Glance

Component	Purpose	When You Need It
`langchain-core`	Base interfaces (Runnables, Messages, Tools)	Always (installed automatically)
`langchain`	High-level chains, agents, retrieval	Most applications
`langgraph`	Graph-based agent orchestration	Complex agent workflows
`langsmith`	Tracing, evaluation, monitoring	Testing & production
`langchain-openai`	OpenAI integration	If using OpenAI models
`langchain-anthropic`	Anthropic integration	If using Claude models
`langchain-community`	700+ third-party integrations	As needed

[Docs: docs.langchain.com/oss/python/concepts/architecture]

Installation

Basic Installation

```
# Core + high-level features
pip install langchain

# Add provider packages as needed
pip install langchain-openai        # OpenAI models
pip install langchain-anthropic     # Claude models
pip install langchain-google-genai  # Gemini models

# Optional: orchestration & observability
pip install langgraph
pip install langsmith
```

Version Note

LangChain v1.2.x is the current stable release. The legacy `langchain-classic` package contains deprecated chains and is **not recommended** for new projects.

Environment Setup

API Key Configuration

```
# Option 1: Environment variables (recommended)
export OPENAI_API_KEY="sk-..."
export ANTHROPIC_API_KEY="sk-ant-..."
export LANGSMITH_API_KEY="lsv2_..."  # optional

# Option 2: Pass directly in code
from langchain_openai import ChatOpenAI
llm = ChatOpenAI(api_key="sk-...")
```

Best Practice

Use environment variables or a `.env` file with `python-dotenv`. Never hardcode API keys in source code.

[Docs: docs.langchain.com/oss/python/getting-started/installation]

Your First LangChain Program

Hello LangChain

```python
from langchain_openai import ChatOpenAI
from langchain_core.messages import HumanMessage

# 1. Initialize a chat model
llm = ChatOpenAI(model="gpt-4o")

# 2. Send a message
response = llm.invoke([
    HumanMessage(content="What is LangChain in one sentence?")
])

# 3. Print the result
print(response.content)
# => "LangChain is a framework for building applications
#     powered by large language models."
```

Key observation: The `invoke()` method is the universal interface. Every LangChain component supports it.

Package Architecture Deep Dive

Your Application

langchain — Agents, Chains, Retrieval helpers

langgraph — Graph orchestration, state management, persistence

langchain-core — Runnables, Messages, PromptTemplates, Tools, OutputParsers

Provider Packages — langchain-openai, langchain-anthropic, langchain-community, ...

Core Abstractions Preview

Abstraction	What It Does
Runnable	Universal protocol: anything with `invoke()`, `batch()`, `stream()`
ChatModel	Wraps an LLM provider; takes messages, returns messages
PromptTemplate	Formats variables into prompts
OutputParser	Converts raw LLM text into structured data
Tool	A function the LLM can decide to call
Retriever	Fetches relevant documents given a query
Chain (LCEL)	A pipeline of Runnables composed with |
Agent	An LLM that plans and executes tool calls in a loop

Each of these is covered in depth in subsequent sections.

The Runnable Protocol

The Universal Interface

Every component in LangChain implements the **Runnable** protocol, which provides three methods:

- `invoke(input)` — Process a single input
- `batch([inputs])` — Process multiple inputs in parallel
- `stream(input)` — Stream output token by token

Plus async variants: `ainvoke()`, `abatch()`, `astream()`

Why This Matters

Because every component shares this interface, you can **compose** them freely with the pipe operator (|). A prompt template, a model, and a parser all "plug in" the same way.

[Docs: docs.langchain.com/oss/python/concepts/runnables]

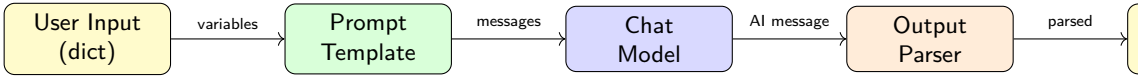
LCEL: LangChain Expression Language (Preview)

Composing with the Pipe Operator

```python
from langchain_openai import ChatOpenAI
from langchain_core.prompts import ChatPromptTemplate
from langchain_core.output_parsers import StrOutputParser

# Define components
prompt = ChatPromptTemplate.from_template(
    "Explain {topic} in simple terms."
)
model = ChatOpenAI(model="gpt-4o")
parser = StrOutputParser()

# Compose into a chain with |
chain = prompt | model | parser

# Use it
result = chain.invoke({"topic": "quantum computing"})
print(result)
```

Full LCEL coverage in Section 03.

Data Flow: How a Request Moves Through LangChain



Each Arrow is a Runnable Boundary

At each step, the output of one Runnable becomes the input of the next. LangChain's tracing (LangSmith) captures every boundary, making debugging straightforward.

[Docs: docs.langchain.com/oss/python/concepts/runnables]

LangChain vs Alternatives

Feature	LangChain	LlamaIndex	Haystack	Raw SDK
Agent framework	✓✓	✓	✓	—
RAG pipeline	✓✓	✓✓✓	✓✓	Manual
Provider agnostic	✓✓✓	✓✓	✓✓	—
Graph orchestration	✓✓✓(LangGraph)	—	—	—
Observability	✓✓✓(LangSmith)	✓	✓	Manual
Community/ecosystem	✓✓✓	✓✓	✓	N/A
Learning curve	Medium	Medium	Medium	Low

Choosing LangChain

LangChain excels when you need **agent orchestration**, **provider flexibility**, and **production observability**. LlamaIndex may be better for pure RAG-focused applications.

What Changed: LangChain v0.x → v1.x

Breaking Changes in v1.0+

- **LCEL is the standard** — Legacy chain classes (`LLMChain`, `SequentialChain`) moved to `langchain-classic`
- **Provider packages split out** — `from langchain.chat_models` → `from langchain_openai`
- **langchain-core extracted** — Interfaces live in `langchain-core`, not `langchain`
- **Agents built on LangGraph** — `create_react_agent` returns a LangGraph graph
- **Python 3.10+ required** — Dropped 3.8/3.9 support

Migration Tip

If migrating from v0.x, start by updating imports to use provider-specific packages and replacing legacy chains with LCEL pipelines.

The Learning Path Ahead

Foundation: Models, Prompts, Parsing, LCEL (Sections 01–03)

↓

Interaction: Tools, Agents, Memory (Sections 04–06)

↓

Retrieval: Documents, Embeddings, RAG (Sections 07–09)

↓

Observability: Callbacks, Streaming, Tracing (Section 10)

↓

Advanced: LangGraph, Deep Agents (Sections 11–12)

↓

Integration: Full Project, Production Practices (Sections 13–14)

Summary

Key Takeaways

1. LangChain is a **framework for composing LLM-powered applications**, not an LLM itself
2. The ecosystem spans **four pillars**: langchain-core (interfaces), langchain (features), LangGraph (orchestration), LangSmith (observability)
3. Everything implements the **Runnable protocol**: `invoke()`, `batch()`, `stream()`
4. Components compose with the **pipe operator** (|) via LCEL
5. Provider packages are **separate**: install only what you need
6. v1.x is a **major redesign** — use LCEL, not legacy chains

Cheat Sheet: Essential Imports

Imports You'll Use Most Often

```python
# Core
from langchain_core.messages import HumanMessage, AIMessage
from langchain_core.prompts import ChatPromptTemplate
from langchain_core.output_parsers import StrOutputParser
from langchain_core.runnables import RunnablePassthrough

# Models (pick your provider)
from langchain_openai import ChatOpenAI
from langchain_anthropic import ChatAnthropic

# Agents & Tools
from langchain.agents import create_react_agent
from langchain_core.tools import tool

# RAG
from langchain_community.document_loaders import PyPDFLoader
from langchain_text_splitters import RecursiveCharacterTextSplitter
from langchain_openai import OpenAIEmbeddings
from langchain_community.vectorstores import FAISS
```

Self-Check Questions

Test your understanding of this section:

1. What problem does LangChain solve that raw LLM SDKs do not?
2. Name the four pillars of the LangChain ecosystem and their roles.
3. What is the Runnable protocol, and why does it matter for composition?
4. Why are LLM provider packages (e.g., `langchain-openai`) separate from the core `langchain` package?
5. What is LCEL and what operator does it use to compose components?
6. How does LangChain v1.x differ from v0.x in terms of chain construction?
7. When might you choose **not** to use LangChain?

Answers are covered in the slides above. Review any you're unsure about before proceeding to Section 01.